

Gated Boost Ensemble of INP-Former and CPR for Class-Agnostic Anomaly Segmentation

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Abstract

We present **GBE-AD** (Gated Boost Ensemble for Anomaly Detection), an ensemble approach for anomaly detection and segmentation on the MVTec AD 2 dataset, combining INP-Former and Cascade Patch Retrieval (CPR). INP-Former leverages Intrinsic Normal Prototypes with a frozen DINOv2 encoder to reconstruct normal features, while CPR uses cascade patch matching for fine-grained anomaly localization. Our key contribution is a gated boosting ensemble strategy where CPR selectively amplifies INP-Former’s anomaly signal only in regions where CPR exhibits high confidence. The CPR contribution weight per category is automatically determined by the edge correlation between the two models’ heatmaps on validation data, requiring no manual tuning. We apply guided filtering for edge-aware smoothing and use Extreme Value Theory (EVT) for principled threshold selection, fitting a GEV distribution on combined validation scores. Both models are trained with synthetic lighting augmentation to improve robustness to the illumination shifts present in MVTec AD 2. On the public test set, our method achieves a mean SegF1 of 0.407.

1. Introduction

1.1. Background

Industrial anomaly detection requires identifying and localizing defects in manufactured products. The MVTec AD 2 dataset introduces real-world challenges including subtle lighting variations between training and test images, requiring models robust to distribution shifts. The pixel-level F1 score (SegF1) evaluation demands both accurate anomaly localization and effective threshold selection.

1.2. Challenge Description

The VAND 2026 Industrial Track evaluates anomaly segmentation performance on 8 product categories (can, fabric, fruit_jelly, rice, sheet_metal, vial, wallplugs, walnuts) under both regular and mixed lighting conditions. Models must be class-agnostic, with a unified pipeline that adapts automatically across categories.

1.3. Related Work

The winning solution of the VAND 2025 Challenge (Category 1), ISVL [1], also combined INP-Former and CPR with a tiling strategy for high-resolution images and morphological post-processing. However, their approach employed per-category model selection, choosing different architectures for different object categories. In contrast, our method is fully class-agnostic, the same pipeline and automatically derived parameters are applied uniformly across all categories. We build upon their insight that INP-Former and CPR are complementary, but introduce a principled ensemble weighting mechanism based on edge correlation that eliminates the need for manual per-category tuning.

2. Methodology

We introduce **GBE-AD** (Gated Boost Ensemble for Anomaly Detection), a class-agnostic ensemble framework for anomaly segmentation.

2.1. Model Design

2.1.1. Approach

We employ a two-model ensemble combining complementary anomaly detection paradigms:

1. **INP-Former** [4]: A reconstruction-based approach using Intrinsic Normal Prototypes that learns to reconstruct normal features directly from test images.
2. **CPR (Cascade Patch Retrieval)** [5]: A retrieval-based approach that matches test patches against a memory bank of normal training patches.

The rationale is that reconstruction and retrieval methods have different failure modes — INP-Former captures global structural anomalies while CPR provides sharper local patch-level detection.

2.1.2. Architecture

INP-Former uses a frozen DINOv2 ViT-Base encoder [6] (dinov2reg_vit_base_14) with learned Intrinsic Normal Prototype tokens. Since MVTEC AD 2 images are high-resolution (e.g., 2232×1024), directly resizing to the model’s input size (448×448) would lose fine-grained defect details. We instead employ an overlapping tiling strategy: each image is divided into an adaptive grid of tiles (target size 600px, overlap 20%), with edges mirror-padded by 10% of the tile size to provide context at boundaries. Each tile is processed independently by the model, and the resulting anomaly maps are stitched back by cropping the padded margins and averaging overlapping regions. This preserves spatial resolution while keeping each tile within the model’s effective receptive field.

CPR uses a frozen DenseNet201 backbone for feature extraction, followed by foreground estimation (LDA-based), global retrieval (K-means + histogram similarity), and local patch matching with learned descriptors trained on synthetic anomaly data.

2.1.3. Ensemble: Gated Boost with Edge Correlation Weighting

Rather than a simple weighted average of the two models’ anomaly maps, we propose a gated boosting strategy where CPR selectively amplifies INP-Former’s signal. The combined anomaly score for each pixel is:

$$\text{combined} = \text{INP} \cdot (1 + w_{\text{cpr}} \cdot \text{CPR}_{\text{clipped}}^{1.5} \cdot G) \quad (1)$$

where INP and CPR are the normalized anomaly scores from each model, $\text{CPR}_{\text{clipped}} = \max(0, \text{CPR})$, and G is a binary gate that equals 1 only where CPR’s score exceeds its own 95th percentile for that image, and 0 otherwise. This gate ensures that CPR only boosts regions where it has high confidence, preventing noisy low-scoring CPR pixels from affecting the output. The exponent of 1.5 further sharpens the CPR signal, suppressing weak activations while amplifying strong detections.

Automatic CPR Weight via Edge Correlation. The per-category weight w_{cpr} determines how much influence CPR has for a given product category. We derive it automatically from validation data by measuring the *edge correlation* between the two models’ heatmaps. Specifically, for each validation image we compute gradient magnitudes (via Sobel filtering) of both the INP-Former and CPR heatmaps, then compute the Pearson correlation between these gradient maps. The mean correlation across all validation images for a category directly serves as w_{cpr} .

The intuition is straightforward: when both models detect boundaries in the same locations (high edge correlation), CPR provides reliable complementary information and should be given more weight. When their edges disagree (low correlation), CPR would introduce noise rather than useful signal. This approach requires no manual tuning — the weight is derived entirely from the models’ agreement on validation data. Table 1 shows the resulting weights.

Category	Edge Correlation	w_{cpr}
vial	0.539	0.539
fruit_jelly	0.312	0.312
can	0.258	0.258
walnuts	0.097	0.097
fabric	0.065	0.065
sheet_metal	0.023	0.023
rice	0.018	0.018
wallplugs	0.018	0.018

Table 1. Per-category CPR weights derived from edge correlation on validation data.

2.2. Post-Processing

After combining the anomaly maps, we apply an edge-aware guided filter [3] (radius=8, $\epsilon=0.001$) using the original test image as a guide. The guided filter smooths homogeneous regions of the heatmap to reduce spurious noise while preserving sharp transitions at object boundaries, where anomalies are most likely to occur. Unlike Gaussian smoothing, which blurs indiscriminately, the guided filter adapts its smoothing kernel to the structure of the input image, ensuring that anomaly edges remain aligned with actual visual boundaries.

2.3. Threshold Selection

Extreme Value Theory (EVT). We fit a Generalized Extreme Value (GEV) distribution on the combined validation heatmap scores (top 5% tail) per category. At test time, pixels with p-value below the false discovery rate (FDR=0.05) are classified as anomalous. This provides a principled, data-driven threshold without requiring labeled anomaly examples.

Normalization. Both models’ heatmaps are normalized using global min-max statistics computed across all validation images before combining.

2.4. Training

INP-Former: Trained for 25 epochs per category with overlapping tiling (target_tile=600), input size 448, crop size 392, batch size 16. Synthetic lighting augmentation applied with probability 0.7, intensity range [0.02, 0.12], stacking up to 2 augmentations per image. Lighting modes include directional gradients, over/underexposure, color tint, and warmth shifts.

CPR: Preprocessing includes foreground mask generation, global retrieval index construction, and synthetic anomaly data generation (6000 samples per category using Perlin noise + DTD textures). Trained for 3000 steps per category with contrastive loss, with the same lighting augmentation applied during training.

2.5. Dataset & Evaluation

Benchmark Dataset Utilization. We use the MVTec AD 2 [2] training images (regular lighting only) for model training and the validation split (normal images only) for threshold calibration. Validation images are used without augmentation for EVT fitting, as experiments showed augmented validation provided no benefit. The test_public split with ground truth masks is used for development evaluation.

Evaluation Criteria. Performance is measured using pixel-level SegF1 with micro-averaging across all pixels per category, following the official MVTec AD 2 evaluation protocol. Precision and recall are computed over the complete pixel set, requiring selection of a single decision threshold.

Evaluation Setting. Our method follows the regular (non-zero-shot) setting. Both models are trained on the MVTec AD 2 training set and use the training data for feature extraction and retrieval index construction. The pipeline is fully class-agnostic — all hyperparameters and the CPR weighting are determined automatically from validation data without category-specific manual tuning.

3. Results

3.1. Performance Metrics

Table 2 shows per-category results on the test_public split.

3.2. Key Findings

- The gated boost ensemble with edge correlation weighting improved mean SegF1 by +0.045 over INP-Former alone, primarily through improvements on fruit_jelly (+0.35) and vial (+0.06).
- Categories with high edge correlation between models (vial, fruit_jelly) benefit significantly from the ensemble,

Category	SegF1	Precision	Recall
can	0.003	0.001	0.090
fabric	0.223	0.349	0.170
fruit_jelly	0.628	0.811	0.512
rice	0.671	0.686	0.656
sheet_metal	0.432	0.653	0.323
vial	0.389	0.597	0.289
wallplugs	0.220	0.206	0.237
walnuts	0.686	0.689	0.683
Mean	0.407	0.499	0.370

Table 2. Per-category SegF1, precision, and recall on MVTec AD 2 test_public.

while categories with low agreement effectively use only INP-Former.

- Guided filtering improved precision by preserving edge-aligned anomaly boundaries.
- EVT-based thresholding with FDR=0.05 provided the best precision-recall balance.

3.3. Ablation Study

Table 3 shows the contribution of each component, starting from a naive ensemble baseline.

4. Discussion

4.1. Challenges & Solutions

- **Lighting robustness:** MVTec AD 2 test images include subtle lighting variations not present in training. We addressed this through synthetic lighting augmentation during training. However, some categories (especially can) remain challenging, possibly due to highly reflective surfaces.
- **Threshold selection:** Without labeled anomaly validation data, finding the optimal threshold is inherently difficult. EVT provides a principled approach but relies on the assumption that validation normal scores represent the true normal distribution.
- **Class-agnostic ensembling:** Different categories benefit differently from CPR. Our edge correlation metric automatically identifies beneficial categories without manual tuning.

4.2. Model Robustness & Adaptability

The gated boost mechanism provides inherent robustness — CPR noise is filtered by the 95th percentile gate, and edge correlation weighting suppresses CPR for categories where it disagrees with INP-Former. The lighting augmentation during training improves robustness to the illumination distribution shifts in the test set.

Configuration	SegF1	P	R
Naive ensemble (average, equal weights)	0.231	0.258	0.466
+ INP-Former only (no CPR)	0.316	0.326	0.415
+ Gated boost + EdgeCorr weighting	0.336	0.328	0.443
+ Guided filter (full GBE-AD)	0.407	0.499	0.370

Table 3. Ablation study showing the contribution of each component on test-public.

4.3. Future Work

- Investigating stronger lighting augmentation or test-time augmentation for better robustness.
- Exploring per-category adaptive thresholds using validation score distributions.
- Training with more epochs and different tiling configurations for improved base model performance.
- Incorporating attention-based ensemble weighting that operates at the pixel level rather than category level.

5. Conclusion

We presented an ensemble anomaly detection method combining INP-Former and CPR with a novel gated boost strategy. The key innovation is using edge correlation between model heatmaps on validation data to automatically determine per-category ensemble weights, achieving a class-agnostic pipeline. Our method achieves a mean SegF1 of 0.407 on the MVTec AD 2 public test set, with strong performance on rice (0.671), walnuts (0.686), and fruit_jelly (0.628). The approach demonstrates that complementary model paradigms can be effectively combined when their agreement is properly measured and leveraged.

References

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